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$$w = \sqrt{u^2 + v^2 + z^2}, u = 3e^t \sin s, v = 3e^t \cos s, z = 4e^t$$

$$\frac{\partial w}{\partial s} = \left(\frac{u}{\sqrt{u^2 + v^2 + z^2}} \quad \frac{v}{\sqrt{u^2 + v^2 + z^2}} \quad \frac{z}{\sqrt{u^2 + v^2 + z^2}} \right) \cdot \begin{pmatrix} 3e^t \cos s \\ -3e^t \sin s \\ 0 \end{pmatrix}$$

$$\frac{\partial w}{\partial s} = \frac{0}{\sqrt{u^2 + v^2 + z^2}} = 0$$

$$\frac{\partial w}{\partial t} = \left(\frac{u}{\sqrt{u^2 + v^2 + z^2}} \quad \frac{v}{\sqrt{u^2 + v^2 + z^2}} \quad \frac{z}{\sqrt{u^2 + v^2 + z^2}} \right) \cdot \begin{pmatrix} 3e^t \sin s \\ 3e^t \cos s \\ 4e^t \end{pmatrix}$$

$$\frac{\partial w}{\partial t} = \frac{25e^{2t}}{5e^t} = 5e^t$$

References